Supplementary Information

Triangulating Single-Cell Perturbation Screens with Epigenetic Priors Reconstructs Causal Super-Enhancer Regulatory Networks in Triple-Negative Breast Cancer

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Supplementary Note 1: Mathematical Foundations of Causal Triangulation & Composite Scoring

In observational single-cell RNA sequencing, technical dropouts (excess zero counts) induce spurious conditional dependencies between independent genes, producing high false discovery rates (>35–45%) when applying standard relevance networks, Gaussian graphical models, or observational tree ensembles (GENIE3, GRNBoost2). To overcome this observational confounding, we formulate a tripartite causal triangulation framework that unites physical chromatin-binding priors from ENCODE ChIP-seq with functional interventional evidence from single-cell CRISPRi.

Let i denote a candidate transcription factor, e denote an accessible enhancer locus, and j denote a downstream target gene. The triangulated causal regulatory weight $W_{causal}(i \rightarrow e \rightarrow j)$ is formulated as the composite product of physical binding confidence and functional perturbation responsiveness:

$$W_{causal}(i \rightarrow e \rightarrow j) = C_{ChIP}(i, e) \times |\log2\ FC(e, j)| \times |t(e, j)|$$

where $C_{ChIP}(i, e)$ represents the peak binding confidence score from ENCODE Factorbook/UCSC clusters normalized to [0, 1000]; log2 FC(e, j) is the interventional expression shift following CRISPRi enhancer silencing; and t(e, j) is the Welch t-statistic measuring perturbation significance against non-targeting control (NTC) sgRNAs. Multiplying |log2 FC| by |t| quadratically rewards effect size while inversely weighting by measurement variance, ensuring that high-weight edges possess both large biological fold change and high statistical precision.

Supplementary Note 2: Directed Acyclic Graph (DAG) Identifiability & Pearl's Do-Calculus

Traditional observational GRN algorithms can only identify graphs up to their Markov Equivalence Class (represented as Completed Partially Directed Acyclic Graphs, CPDAGs), leaving edge directions reversible in v-structures without randomized interventions. Under Judea Pearl's structural causal model (SCM) framework, experimental CRISPRi perturbations act as exogenous, randomized atomic interventions:

$$P(Y_j \mid do(E_e = 0)) \neq P(Y_j \mid E_e = 0)$$

Because the guide RNA targets enhancer locus e independently of all endogenous cellular confounding (cell cycle, batch effects, transcriptional bursting), the perturbation breaks all incoming back-door paths to e. Consequently, any statistically significant shift in gene j must be downstream of e, unambiguously establishing the directed functional edge e → j.

Importantly, a rigorous theoretical distinction must be acknowledged: while the functional regulatory arm (e → j) is strictly identified through randomized experimental intervention (do(e = 0)), the upstream physical binding arm (i → e) represents an observational chromatin occupancy prior derived from ChIP-seq. Triangulation integrates these distinct evidentiary streams by prioritizing cascades that are simultaneously physically plausible and functionally responsive under intervention, establishing defensible directed regulatory flow without claiming direct genetic perturbation of the upstream transcription factor coding genes themselves.

Supplementary Note 3: Fast Vectorized Two-Sample Welch t-Test for Genome-Scale Perturb-seq

Evaluating 11,975 individual sgRNA guides across 95,678 single cells and 32,799 gene features requires computing millions of hypothesis tests. To avoid computationally prohibitive nested loops, we implemented a vectorized two-sample Welch t-test:

$$t = (\bar{X}_1 - \bar{X}_0) / \sqrt{((s_1^2 / n_1) + (s_0^2 / n_0))}$$

The effective degrees of freedom (ν) are computed via the Welch-Satterthwaite equation:

$$\nu \approx [(s_1^2 / n_1) + (s_0^2 / n_0)]^2 / [(s_1^2 / n_1)^2 / (n_1 - 1) + (s_0^2 / n_0)^2 / (n_0 - 1)]$$

To eliminate confounding from combinatorial epistatic interactions between co-expressed guides, the differential perturbation analysis was computed strictly using the single-guide cohort (n = 3,930 cells) contrasted against the unassigned negative control pool (n = 7,767 cells). Significant interventional responses were defined under stringent dual thresholds: $|\log2\ FC| \geq 0.25$ and $|t| \geq 2.5$ (corresponding to two-sided $p < 0.012$).

Supplementary Note 4: Directed Multigraph Topology & Google PageRank Prioritization

To identify which transcription factors exert master regulatory control over the global network architecture, we mapped the triangulated causal edges into a directed weighted multigraph G = (V, E, W). We implemented Google PageRank with a damping factor α = 0.85 and maximum 500 iterations (convergence tolerance 1×10^{-6}):

$$PR(u) = (1 - \alpha) / N + \alpha \sum_{v \in In(u)} [PR(v) / L(v)]$$

Nodes were simultaneously evaluated by cumulative outgoing causal paths, total target genes regulated, and the composite Master Regulatory Score (MRS):

$$MRS(i) = \beta \times TargetPercentile(i) + (1 - \beta) \times WeightPercentile(i)$$

where β = 0.5 balances broad target reach with deep biochemical intensity. CTCF, POLR2A, EP300, JUND, and MYC consistently ranked in the top percentiles across all topological metrics.

Supplementary Note 5: Explainable AI (SHAP TreeExplainer) & In Silico Driver Knockout Validation

To capture non-linear, multi-factorial causal gene regulation beyond monotonic assumptions, we deployed XGBoost Regressors predicting target gene responsiveness from master regulator causal weights. Exact feature attribution values were computed via the TreeExplainer algorithm:

$$\varphi_i(x) = \sum_{S \subseteq F \setminus \{i\}} [|S|! (|F| - |S| - 1)! / |F|!] \times [f_x(S \cup \{i\}) - f_x(S)]$$

TreeExplainer SHAP attribution identified JUND (mean |SHAP| = 0.0296), MYC (mean |SHAP| = 0.0174), MAX (mean |SHAP| = 0.0125), and EP300 (mean |SHAP| = 0.0117) as top non-linear drivers.

Feature dependency analysis for JUND uncovered switch-like threshold dynamics: causal weights below 8.5 conferred minimal impact, whereas weights exceeding 10.0 induced rapid, non-monotonic transactivation.

In silico driver knockout simulations were performed across N = 864 target genes by numerically clamping driver causal weights to zero (do(TF = 0)) and evaluating model expression shifts against experimental CRISPRi log2 fold changes. This empirical validation yielded Pearson $r = 0.688$ and $R^2 = 0.473$ ($p = 3.37 \times 10^{-122}$), confirming high predictive fidelity.

Supplementary Note 6: Biochemical Mechanism of Synergistic Epigenetic Combination Therapies

A fundamental biological rationale underlies the predicted therapeutic synergy of combining p300 HAT inhibitors (A-485, CCS1477) with class I HDAC inhibitors (Entinostat, Vorinostat) and BET bromodomain inhibitors (OTX015/Birabresib) in TNBC, despite their opposing biochemical effects on bulk histone acetylation:

1. Targeted Oncogenic Super-Enhancer Collapse: In basal-like TNBC, p300 HAT specifically acetylates histone H3K27 at hyper-dense super-enhancers driving core oncogenic drivers (MYC, AP-1/JUND). Pharmacological p300 inhibition selectively dismantles these multi-transcription-factor condensates, depriving oncogenes of transcriptional burst frequency.

2. Tumor Suppressor and Pro-Apoptotic Gene Reactivation: In parallel, aberrant class I HDAC activity in TNBC silences endogenous tumor suppressor pathways and apoptotic effectors (CDKN1A/p21, BIM, PTEN). Selective HDAC inhibition relieves this transcriptional repression, reactivating cell cycle checkpoints and apoptotic cascades.

3. Epigenetic Plasticity Abrogation: Combining super-enhancer disruption (BETi + HATi) with tumor suppressor de-repression (HDACi) simultaneously starves the cell of growth signals while enforcing apoptotic execution, explaining the predicted 93% network collapse in silico.

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